Individual Work Log

PROJECT NAME:	Whole-Body Model Predictive Control for Industrial Humanoids
STUDENT NAME:	**Rumeth Gunasinghe**
STUDENT ID:	**104498227**

Cumulative Summary of week 5

Week 3	Major work done this week	Total hours spent this week	Cumulative Total
1	**Studied PID control fundamentals**	**3**	**2**
2	**Started inverted pendulum simulation implementation**	**4**	**4**
3	**Understanding humanoid robotics and MPC concept**	**2**	**8**
4	**Tested simulation behaviour**	**2**	**6**
5	**Further research on MPC implementation methods**	**2**	**7**
6	**Team meeting and sprint progress discussion**	**1.5**	**3**
7	**Worklog and documentation updates**	**1.5**	**5**
8	**Building the team collab tool (jira)**	**2**	**3**
9			
10			
11			
12			

Week 1

WEEK #	1		
Dates covered:	**02/03/2026 – 08/03/2026**		
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
Researching on potential projects	Completed	**10**	**Asking for personal project proposal**
Communicating with the convenor	completed	**4**	**Choosing a given project from the list**
TOTAL WEEKLY TIME SPENT		**14**	

Summary/weekly reflection for Week 1:

- Researched on available projects to find the suitable one
- Communicating with the convenor for support

Plan for the next week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION
More research into available projects	15/03/2026
Chosing a suitable project	15/03/2026

Week 2

WEEK #	2		
Dates covered:	**09/03/2026 – 15/03/2026**		
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
Researching on projects	Completed	**8**	**Chosing the humanoid robot control project**
Researching on the selected project	Completed	**12**	**Drafting a plan**
TOTAL WEEKLY TIME SPENT		**20**	

Summary/weekly reflection for Week 2:

- Researching on available projects
- Researching on the selected project

Plan for the next week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION
Finding a Teammate	22/03/2026
Finishing up the project and team plan assignment	22/03/2026

Week 3

WEEK #	3		
Dates covered:	**16/-3/2026 – 22/03/2026**		
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
Project allocation and understanding project requirements	Completed	**2**	**Task devision among members**
Research on simulation tools (PyBullet / MuJoCo)	Completed	**5**	**Installed necessary libraries**
Finshing assignment 1	Completed	8	Submitted the assignment
TOTAL WEEKLY TIME SPENT		**15**	

Summary/weekly reflection for Week 3:

- Project was assigned and understood
- Team roles and responsibilities were defined
- Initial research on humanoid robotics and control systems conducted
- Completed major sections of the project proposal/report
- Set up tools for development (GitHub, communication platforms)

Plan for the next week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION
Working on Assignment 2 (research report)	29/03/2026
Researching and start working on the project implementation	On going

WEEK #	4		
Dates covered:	**23/03/2026 – 29/03/2026**		
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
Completed research document	Completed	5	Research doc submission
Continued research on Model Predictive Control (MPC)	On going	**4**	drafts
Research on inverted pendulum model	On going	4	drafts
Planning system architecture	Completed	**3**	Finalised System Architecture
TOTAL WEEKLY TIME SPENT		**16**	

Summary/weekly reflection for Week #:

- Conducted in-depth research on MPC and control systems
- Identified inverted pendulum as the correct modelling approach
- Defined high-level system architecture
- Started research report writing
- Set up Jira sprint structure

Plan for the next week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION
Begin implementation of inverted pendulum simulation	05/04/2026
Testing simulation Behavior	05/04/2026

WEEK #	5		
Dates covered:			
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
Studied PID control fundamentals	Completed	3	Understood feedback control and tuning parameters
Started inverted pendulum simulation implementation	Completed	4	Implemented basic system model in Python
Tested simulation behaviour	Completed	2	Verified system response under simple conditions
Further research on MPC implementation methods	Completed	2	Focused on optimisation-based control
TOTAL WEEKLY TIME SPENT		11	

Summary/weekly reflection for Week #:

- *Studied PID control as baseline method*
- *Started implementation of inverted pendulum simulation*
- *Conducted initial testing of system behaviour*
- Difference between reactive (PID) and predictive (MPC) control
- Importance of feedback loops in maintaining system stability
- Simulation helps validate concepts before complex implementation

Plan for the next week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION
Working on Sprint report 1	19/04/2026
Implementing and testing on PID and MPC	19/04/2026

WEEK #	6		
Dates covered:			
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
	Completed		
	Not started		
	On hold		Check with xyz
TOTAL WEEKLY TIME SPENT			

Summary/weekly reflection for Week #:

-key tasks done / things attended

-key things learned about Computing Technology projects

-any literature read and key things learned

-Issues/problems/Challenges

Plan for the next week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION

WEEK #	7		
Dates covered:			
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
	Completed		
	Not started		
	On hold		Check with xyz
TOTAL WEEKLY TIME SPENT			

Summary/weekly reflection for Week #:

-key tasks done / things attended

-key things learned about Computing Technology projects

-any literature read and key things learned

-Issues/problems/Challenges

Plan for the next week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION

WEEK #	8		
Dates covered:			
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
	Completed		
	Not started		
	On hold		Check with xyz
TOTAL WEEKLY TIME SPENT			

Summary/weekly reflection for Week #:

-key tasks done / things attended

-key things learned about Computing Technology projects

-any literature read and key things learned

-Issues/problems/Challenges

Plan for the next week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION

WEEK #	9		
Dates covered:			
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
	Completed		
	Not started		
	On hold		Check with xyz
TOTAL WEEKLY TIME SPENT			

Summary/weekly reflection for Week #:

-key tasks done / things attended

-key things learned about Computing Technology projects

-any literature read and key things learned

-Issues/problems/Challenges

Plan for the next week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION

WEEK #	10		
Dates covered:			
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
	Completed		
	Not started		
	On hold		Check with xyz
TOTAL WEEKLY TIME SPENT			

Summary/weekly reflection for Week #:

-key tasks done / things attended

-key things learned about Computing Technology projects

-any literature read and key things learned

-Issues/problems/Challenges

Plan for the next week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION

WEEK #	11		
Dates covered:			
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
	Completed		
	Not started		
	On hold		Check with xyz
TOTAL WEEKLY TIME SPENT			

Summary/weekly reflection for Week #:

-key tasks done / things attended

-key things learned about Computing Technology projects

-any literature read and key things learned

-Issues/problems/Challenges

Plan for the next week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION

WEEK #	12		
Dates covered:			
TASKS	**STATUS**	**TIME SPENT**	**ACTION ITEM/NOTE/EVIDENCE**
	Completed		
	Not started		
	On hold		Check with xyz
TOTAL WEEKLY TIME SPENT			

Summary/weekly reflection for Week #:

-key tasks done / things attended

-key things learned about Computing Technology projects

-any literature read and key things learned

-Issues/problems/Challenges

Plan for the next week:

TASKS PLANNED FOR NEXT WEEK	EXPECTED COMPLETION